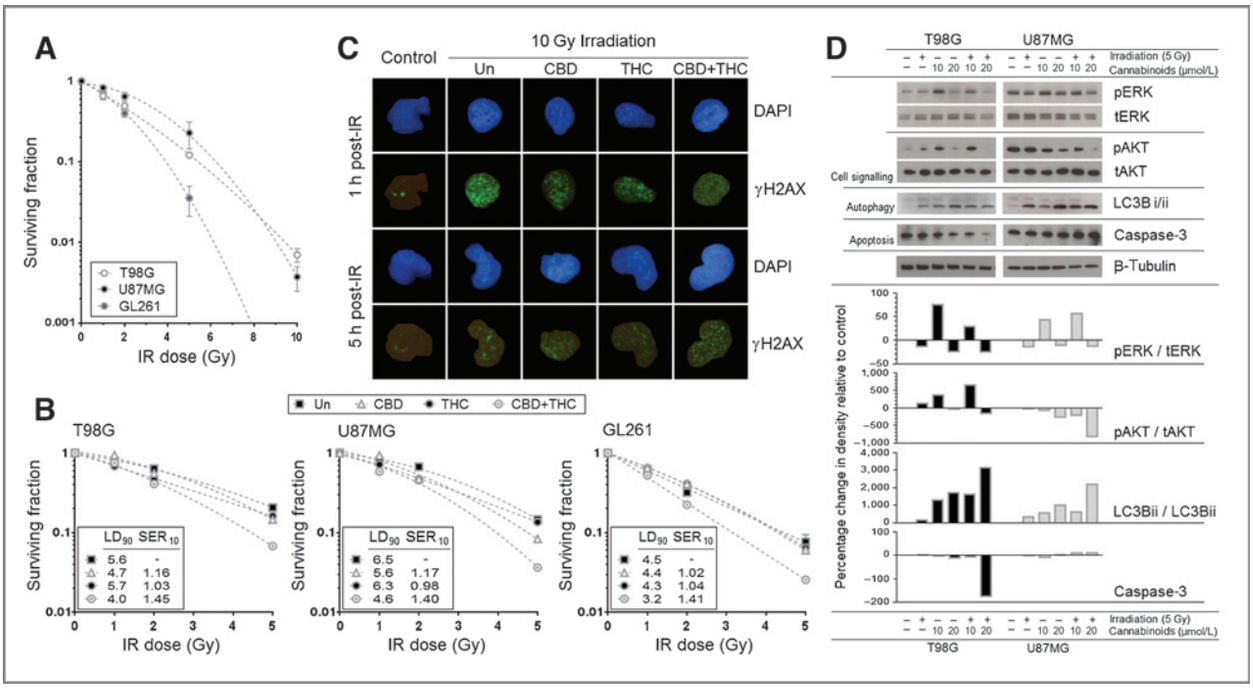

Figure 3. Effect of CBD, THC, and irradiation on colony-forming efficiency. T98G, U87MG, and GL261 were cultured with the pure forms of CBD and THC individually or concomitantly for 4 hours, before irradiation (<10 Gy). Clonogenic potential was then assessed. Irradiation was active in the cell lines, GL261 being most radiosensitive (A). Using either CBD or THC singly before irradiation did not alter the fraction of surviving cells (B); however, preculturing cells with the combination of CBD and THC enhanced the activity of irradiation, as the dose required to kill 90% (LD_{90}) of cells was increased. For example, in T98G cells, LD_{90} was altered from 5.6 Gy to 4.0 Gy (~SER_{10} of 1.45; B). Irradiating cells (10 Gy) resulted in an increase in γ-H2AX foci, a marker of DNA double-strand breaks. Foci persisted for a longer duration in cells pretreated with CBD and THC when compared with irradiated cells that had no pretreatment (C). Western blotting was performed to assess the effects of these combinations on markers of cell signaling, apoptosis, and autophagy (D).

A combination of CBD and THC enhances the effects of radiation

For gliomas that are accessible to it, surgery is the principal treatment. This is commonly supported by radiotherapy and/or chemotherapy, and so for this reason we assessed the benefit of combining the cannabinoids with γ-irradiation. We assessed whether the cannabinoids can prime cells to irradiation and more specifically studied whether in this instance they should be used individually or with each other. Methodologically, cells were pretreated for 4 hours with a single 20 μmol/L dose of CBD-P or THC-P, or with a combination of CBD-P and THC-P (equimolar dose of 10 μmol/L of both), before being subjected to ionizing radiation. Treated cells were then reseeded into plates, and colony-forming efficiency was assessed by enumerating colonies after 14 days using methylene blue.

Results showed that all three cell lines tested were radiosensitive, with a surviving fraction at 5 Gy of 0.12 ± 0.016, 0.22 ± 0.11, and 0.035 ± 0.025 for T98G, U87MG, and GL261, respectively (Fig. 3A). Following pretreatment with single agent CBD or THC, there was no significant difference in colony numbers, but there was, however, a trend toward a smaller surviving fraction when CBD and THC were used in combination (Fig. 3B). For example, in U87MG cells, the extrapolated LD_{90} for radiation alone was 6.5 Gy, and 5.6 Gy and 6.3 Gy when used in combination with CBD or THC. However, this was reduced to 4.6 Gy when cells were pretreated with CBD and THC concomitantly before irradiation, giving an SER_{10} of 1.4 (Fig. 3B). The effect of the cannabinoids on the repair of DNA double-strand breaks caused by irradiation was assessed using γ-H2AX staining, whereby γ-H2AX foci indicate individual sites of damage. 10 Gy caused a substantial increase in the number of these foci, present 1 hour after irradiation, and this increase was unaffected by pretreatment with the cannabinoids. Five hours after irradiation, a large proportion of the foci were repaired and fluorescence intensity in the irradiation alone cells decreased. In the cannabinoid pretreated cells, however, fewer foci were repaired and fluorescence intensity remained higher, particularly in those cells pretreated with a combination of CBD and THC (Fig. 3C). These results suggested that cannabinoids slowed the repair of double-strand breaks and that DNA damage persisted in these cells.

To further understand the mechanisms of the benefit of combining cannabinoids with irradiation, we examined the markers of general signaling pathways as well as

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